

This page is to be omitted in your dissertation!

Note: This template is only an example. It shows what needs to be included for a typical project that includes implementation and system building. Different projects may require different format. Please consult your supervisor for the most suitable format.

The Project Title

Dissertation

COMP702 — M.Sc. Project (2025/26)

Submitted by

Your Name

(Your student number)

under the supervision of

Supervisor's Name

SCHOOL OF COMPUTER SCIENCE AND INFORMATICS
UNIVERSITY OF LIVERPOOL

Student Declaration

I confirm that I have read and understood the University's Academic Integrity Policy.

I confirm that I have acted honestly, ethically and professionally in conduct leading to assessment for the programme of study.

I confirm that I have not copied material from another source nor committed plagiarism nor fabricated data when completing the attached piece of work. I confirm that I have not previously presented the work or part thereof for assessment for another University of Liverpool module. I confirm that I have not copied material from another source, nor colluded with any other student in the preparation and production of this work.

I confirm that I have not incorporated into this assignment material that has been submitted by me or any other person in support of a successful application for a degree of this or any other university or degree-awarding body.

SIGNATURE _____

DATE April 1, 2026

Acknowledgments

You can state your acknowledgement here.

The Dissertation Title

Abstract

Provide some brief description of your project, what the goals were, what experiments were to be performed, what software was to be developed, what was accomplished in the project, etc.

This should provide some overview that could generally be understood by a non-specialist.

Statement of Ethical Compliance

Provide the statement of ethical compliance here.

Contents

1	Introduction	7
1.1	Background and Motivation	7
1.2	Problem Statement	7
1.3	Brief Description of the Approach	7
1.4	Outcome	8
2	Design	9
2.1	The Architecture	9
2.2	Detailed Description of Each Component	9
3	Implementation	10
3.1	Component A	10
3.2	Component B	10
4	Experimental Results	11
4.1	Experiment 1	11
4.2	Experiment 2	11
5	Theoretical Analysis	12
6	Conclusion and Future Work	13
7	BCS Project Criteria and Self Reflection	14
	Bibliography	15
A	An Analysis of the LLM Model on Formal Mathematical Statements	16
B	Another Interesting Discovery	17

List of Figures

List of Tables

Chapter 1

Introduction

This contains the background and main motivation of your project. Here is an example.

1.1 Background and Motivation

As electronic chips are getting smaller and more complex, their production becomes error prone. Experts estimate that those errors cost around 5 trillion dollars each year [3].

The goal of the project is to develop a practical tool that help identifying the bugs in the design phase of chips production. I will leverage various state of the art techniques from AI, theoretical computer science and formal methods to achieve it. To use my tool, the chip designer needs only to input a chip design written in RTL language Verilog [4] and some specification statements written in Plain English. The tool will automatically test whether the design fulfils the specification.

I believe my tool can make the whole process of electronic chip design safer and more efficient, and ultimately saves the world 5 trillion dollars each year.

1.2 Problem Statement

Here is the formal definition of the main problem that I consider in this dissertation.

Input: A chip design written in Verilog and a specification written in plain English.

Task: Determine if the chip design fulfils the specification.

If it does not, the tool will draw the chip diagram and pinpoint the areas in the chip where bugs have likely occurred.

1.3 Brief Description of the Approach

I develop my own Verilog compiler in Java.

To read the specification, I use the latest LLM tool to translate the English language to a formal mathematical statement written in SMT-LIB – the standard language for formal specification [5].

To test chip design against the specification, I use the latest statistical verification technique from [1]. This technique will also identify the area where bugs likely to occur.

When there is a bug, I use the latest graph drawing technique in [2] to draw the whole chip.

1.4 Outcome

My complete code has been published in my github [6]. It works very well, though it is debatable whether my translator (from plain English to SMT-LIB statement) share the same semantics with typical chip designers. My tool can also be accessed in my private url: `my-private-url`.

Chapter 2

Design

In this chapter, we will outline the design of the software that was developed in this project.

2.1 The Architecture

Here you may want to draw your architecture diagrams that visually represent software components, relationships, and system interactions.

The description can be brief.

2.2 Detailed Description of Each Component

Here you may want to give more detailed description of each component and what tool you use in the development.

Chapter 3

Implementation

In this chapter you want to describe you implement your tool. You can merge this chapter with the previous one, if you wish. You may also want to give some snippets of your code that you are most proud of.

3.1 Component A

I implement component A in C++. The result is excellent.

3.2 Component B

Component B is implemented in a mix of Python and Java.

Chapter 4

Experimental Results

Here you can give the details of some experiments that were performed during the course of the project.

4.1 Experiment 1

Some stuff here.

4.2 Experiment 2

Some more stuff here.

Chapter 5

Theoretical Analysis

Here you can give some theoretical analysis of your system.

Chapter 6

Conclusion and Future Work

Chapter 7

BCS Project Criteria and Self Reflection

Bibliography

- [1] A.N. Author. The best paper of all. *Some really good reference* **12**, pp. 100–113, 2003.
- [2] S. omeone. A paper that I wish I’d written. *J. Combinatorics* **34**, pp. 25–78, 1979.
- [3] S. Expert. Some really expert opinions on chip. *A very good chip journal* **12**, pp. 100–113, 1900.
- [4] Verilog. <https://www.verilog.com/>
- [5] SMT-LIB. <https://smt-lib.org/>.
- [6] My github. www-my-git-hub.

Appendix A

An Analysis of the LLM Model on Formal Mathematical Statements

You may want to add some appendices if there is some material that you find interesting, but does not fit well in the dissertation. As an example:

I run the LLM Model M during the initial development of my project. I found that M uses completely different semantics as typical chip designers. Here are some of the experiments.

Appendix B

Another Interesting Discovery

More appendix.